SUMMER TERM 2022
CENTRALLY-MANAGED ONLINE EXAMINATION
ECON0039: ADVANCED MACROECONOMICS

Solution

Time allowance

You have 2 hours to complete this examination, plus an Upload Window of 20 minutes. The Upload Window is for uploading, completing the Cover Sheet and correcting any minor mistakes and should not be used for additional writing time.

If you have been granted SoRA extra time and/ or rest breaks, your individual examination duration will be extended pro-rata and you will also have the 20-minute Upload Window added to your individual duration.

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Page Limit: No page limit

All questions should be answered.

Part A carries 1/3 of the total mark, Part B 1/3 and Part C 1/3. Questions B.1 and B.2 have the same weight.

If you have a query about the examination paper, instructions or rubric, you should complete an AssessmentUCL Query Form. Please note that you will not receive a response during your examination. By submitting this assessment, you are confirming that you have not violated UCL's Assessment Regulations relating to Academic Misconduct contained in Section 9 of Chapter 6 of the Academic Manual.

PART A : Demand and supply shocks in a "UV" model

THIS PROBLEM considers a general equilibrium model in the UV setting of Lecture 4 on Business Cycles. The economy features a representative firm and a representative household.

Technology : The representative firm produces consumption c and investment x using labour according to the technology

$$(c^\sigma + x^\sigma)^{\frac{1}{\sigma}} = A\ell^\alpha$$

with $0 < \alpha < 1$ and $\sigma \geq 1$. ℓ represents labour input.

Preferences : The representative household utility is of the UV form, and is given by

$$U(c, \ell) + V(x, \theta) = \gamma \ln c - B\ell + \theta \ln x,$$

where ℓ is labour.

Markets : All markets are competitive. The representative household owns the firm and receives its profits π. It is assumed that the consumption good is the numéraire (its price is normalized to one), while the price of investment is q and the wage is w.

The household budget constraint (that will always bind) is

$$c + qx \leq w\ell + \pi.$$

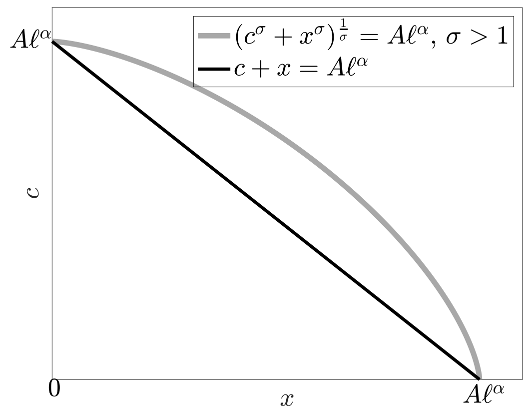
A.1 The profit maximization problem of the firm is

$$\max_{x,c,\ell} \pi = c + qx - w\ell$$

$$\text{subject to } (c^\sigma + x^\sigma)^{\frac{1}{\sigma}} = A\ell^\alpha$$

Figure A1 shows the production possibility frontier (ppf) of the economy for $\sigma = 1$ and $\sigma > 1$. The ppf is the locus of the couples (c, x) that can be produced for a given level of input labour ℓ.

Figure A1 - Production Possibility Frontier for $\sigma = 1$ and $\sigma > 1$



Explain why the slope of the ppf can be interpreted as the negative of the relative price q. Discuss the difference between the case $\sigma = 1$ and $\sigma > 1$.

For given $A\ell^\alpha$, the ppf is given by the equation

$$(c^\sigma + x^\sigma)^{\frac{1}{\sigma}} = A\ell^\alpha$$

The slope s of the ppf in the (x, c) plane can be interpreted as follows: for given total production $A\ell^\alpha$, in order to increase c by one unit, one needs to decrease x by s units. The slope s is therefore the marginal rate of transformation between consumption and investment. With perfectly competitive market, it is also (minus) the relative price of investment in terms of consumption, hence q.

When $\sigma = 1$, we have $c + x = A\ell^\alpha$. This corresponds to a one good economy. The slope of the ppf is constant and equal to -1, and the relative price q is equal to one.

When $\sigma > 1$, we see on Figure A.1 that the relative price q is not constant. When x is close to zero, the slope of the ppf is close to zero, meaning that one needs not to sacrifice a lot of consumption to have more investment. The relative price of investment q is close to zero. On the contrary, q is very large when c goes to zero. In this economy, there is not perfect substitutability between consumption and investment, and shocks will affect the relative price of investment.

A.2 From now on, we assume that σ is large enough, more precisely that $\sigma > \frac{1}{\alpha}$.

Show that profit maximisation can be rewritten as

$$\max_{x,c} \pi = c + qx - wA^{-\frac{1}{\alpha}} (c^\sigma + x^\sigma)^{\frac{1}{\alpha\sigma}}$$

and derive first order conditions to this maximization with respect to c, denoted equation (1), and x (eq. (2)).

The equation

$$(c^\sigma + x^\sigma)^{\frac{1}{\sigma}} = A\ell^\alpha$$

can be rewritten as

$$\ell = A^{-\frac{1}{\alpha}} (c^\sigma + x^\sigma)^{\frac{1}{\alpha\sigma}}.$$

Plug into the definition of profit to obtain

$$\pi = c + qx - wA^{-\frac{1}{\alpha}} (c^\sigma + x^\sigma)^{\frac{1}{\alpha\sigma}}.$$

The first order condition with respect to c is

$$1 = wA^{-\frac{1}{\alpha}} \frac{1}{\alpha} c^{\sigma-1} (c^\sigma + x^\sigma)^{\frac{1}{\alpha\sigma}-1}, \tag{1}$$

and the one with respect to x is

$$q = wA^{-\frac{1}{\alpha}} \frac{1}{\alpha} x^{\sigma-1} (c^\sigma + x^\sigma)^{\frac{1}{\alpha\sigma}-1}. \tag{2}$$

A.3 Show that the utility maximisation problem of the household can be written as

$$\max_{x,\ell} \gamma \ln(w\ell + \pi - qx) - B\ell + \theta \ln x$$

Derive first order conditions to this maximisation, with respect to x (eq. (3)) and ℓ (eq. (4)). The budget constraint is denoted as equation (0).

The budget constraint

$$c + px = w\ell + \pi \tag{0}$$

implies $c = w\ell + \pi - qx$. Replace c in the utility function to obtain

$$U + V = \gamma \ln(w\ell + \pi - qx) - B\ell + \theta \ln x.$$

First order condition with respect to x is

$$\frac{\gamma q}{c} = \frac{\theta}{x}, \tag{3}$$

and the one with respect to ℓ is

$$\frac{\gamma w}{c} = B. \tag{4}$$

A.4 Define the general equilibrium of this economy.

The general equilibrium of the economy is quantities (c, x, ℓ) and prices (q, w) such that *(i)* quantities maximise utility and profits at given prices and *(ii)* prices are such that consumption, investment and labour markets clear.

A.5 Solve for equilibrium prices and quantities.

Hint: (i) use (1) and (2) to find $\frac{x}{c}$ as a function of q (denoted equation (5)); (ii) use (5) and (3) to find q; (iii) derive a solution for $\frac{x}{c}$ (denoted equation (7)); (iv) take (1), rearrrange terms to have $\frac{x}{c}$ and replace it by its solution, use (4) to eliminate w and obtain the equilibrium value of c; (v) solve for x using the solution for $\frac{x}{c}$; (vi) use $(c^\sigma + x^\sigma)^{\frac{1}{\sigma}} = A\ell^\alpha$ to find ℓ.

(i): Take (2)/(1) to obtain

$$\frac{x}{c} = q^{\frac{1}{\sigma - 1}}. \tag{5}$$

(3) can be written

$$\frac{x}{c} = \frac{\theta}{\gamma}\frac{1}{q}. \tag{3'}$$

(ii): (3) and (5) imply

$$\boxed{q = \left(\frac{\theta}{\gamma}\right)^{\frac{\sigma-1}{\sigma}}}. \tag{6}$$

(iii): Therefore (3') implies

$$\boxed{\frac{x}{c} = \left(\frac{\theta}{\gamma}\right)^{\frac{1}{\sigma}}.} \tag{7}$$

(iv): Equation (1) can be written

$$1 = wA^{-\frac{1}{\alpha}}\frac{1}{\alpha}c^{\sigma-1}\left(c^\sigma\frac{(c^\sigma + x^\sigma)}{c^\sigma}\right)^{\frac{1-\alpha\sigma}{\alpha\sigma}}, \tag{1'}$$

which is equivalent to

$$1 = wA^{-\frac{1}{\alpha}}\frac{1}{\alpha}c^{\frac{1-\alpha}{\alpha}}\left(1 + \left(\frac{x}{c}\right)^\sigma\right)^{\frac{1-\alpha\sigma}{\alpha\sigma}}, \tag{1''}$$

Replace $\frac{x}{c}$ using (7) and w using (4) ($w = \frac{Bc}{\gamma}$) to obtain

$$\boxed{c = \alpha^\alpha A\left(\frac{\gamma}{B}\right)^\alpha\left(1 + \frac{\theta}{\gamma}\right)^{\frac{\alpha\sigma-1}{\sigma}}.}$$

(v) Now (7) gives

$$\boxed{x = \alpha^\alpha A\left(\frac{\gamma}{B}\right)^\alpha\left(1 + \frac{\theta}{\gamma}\right)^{\frac{\alpha\sigma-1}{\sigma}}\left(\frac{\theta}{\gamma}\right)^{\frac{1}{\sigma}}.}$$

(vi) Equation $(c^\sigma + x^\sigma)^{\frac{1}{\sigma}} = A\ell^\alpha$ rewrites

$$\ell = A^{-\frac{1}{\alpha}} c^{\frac{1}{\alpha}} \left(1 + \left(\frac{x}{c}\right)^\sigma\right)^{\frac{1}{\alpha\sigma}}$$

Use the solution for c and $\frac{x}{c}$ to obtain

$$\boxed{\ell = \tfrac{\alpha(\gamma+\theta)}{B}.}$$

A.6 Can an investment demand shock ($\partial\theta > 0$) create business cycles comovements? (Don't forget that $\sigma > \frac{1}{\alpha}$.) Explain.

Because $\sigma > \frac{1}{\alpha}$, we have :

	∂c	∂x	∂q	$\partial \ell$
$\partial\theta > 0$	$+$	$+$	$+$	$+$

An investment demand shock creates business cycle comovements: consumption, investment and hours worked increase. The relative price of investment also increases. Therefore, GDP in units of consumption, as defined by $y = c + qx$, also increases.
Why is that so? When $\sigma = 1$ (which is the RBC model we have studied in class), $q = 1$ is constant, and an increase in demand for investment comes at the cost of less consumption. When $\sigma > 1$, because there is not perfect substitution between c and x, part of the increase in x demand (when $\partial\theta > 0$) goes into an increase in q, the relative price of x, so that both c and x increase.

PART B : Questions

WRITE A structured short essay for each of the two themes below. Each essay should be roughly 500 words.

B.1 The fiscal origin of hyperinflation.

- We have studied hyperinflation in Germany in the 1920s and in Zimbabwe in the 2000s. A few observations can be made:
  - hyperinflation happens in periods of political instability,
  - hyperinflation is always related to fiscal deficits being monetised,
  - in anticipation of future inflation, agents run away from money, which reduces the real value of money and further increases inflationary pressures.

- The proximate cause of hyperinflation is always excessive money creation. Why is there at some point excessive money creation? Because the (non independent) central bank prints money to finance government expenditures (including civil servant wages).
- In the Weimar republic, money creation exploded when the government decided to print money to support the workers who went on strike in the Ruhr, following French and Belgian occupation. In Zimbabwe, Mugabe's government financed a war in Democratic Republic of Congo with money creation.
- Once the hyperinflation dynamics is triggered, the economy collapses and there is a decline in the volume of tax collection. Furthermore, because taxes are collected in nominal terms, their real value shrinks in periods of hyperinflation. This is due to the time elapsed between the moment the taxable event occurs and the collection of the tax becomes effective. Hence a doom loop: less tax revenues $\rightsquigarrow$ need for more money creation to cover the budget deficit. This mechanism is known as the *Olivera-Tanzi effect*.
- Therefore, fiscal discipline is always a requirement for an hyperinflation exit plan. This is what we see in the SARGENT model of hyperinflation. In that model, expectations are rational and it is the whole future of the money supply that matters, not the past. A credible plan can therefore abruptly end an hyperinflation episode, as seen in Germany in the 1920's.

B.2 Definition and measure of business cycles.

- Business cycles refer to the successions of expansions and recessions in economic activities, over horizons between (say) 5 and 10 years.
- Business cycles movements imply that most of the economic aggregates move in the same direction (co-movements) and that they show some persistence.
- A reference is the seminal work of BURNS & MITCHELL [1946] *"Measuring Business Cycles"* at the National Bureau of Economic Research.
- There is not a precise and universal definition of it, but it can be defined as the fluctuations of economic activity around its long run trend (which is only precise if we have a definition of the trend).
- The problem of extracting the cycle from a series can be framed as follows. Take a time series $y_t = \log Y_t$. The objective is to decomposed into a "Trend" and a "Cycle" such that

$$y_t = y_t^T + y_t^C$$

  Statistical theory tells us that if y_t is non stationary, there is an infinite number of decomposition into a (stationary) cycle and a (non stationary) trend. Therefore, we need economic judgement to extract the business cycle.
- What are the usual ways of extracting business cycles?

- In the *time domain*, one can take growth rates or remove a linear trend. This is typically leading to a cycle that is too short in the former case and too long in the latter, as compared to the NBER datation. One can use econometric relations to compute *potential output*, and then define the cycle by the distance to potential output, aka the *output gap*. This gives a cycle that is close to the NBER one. Finally, once can use the Hodrick-Prescott filter, that extracts a trend that is flexible but not too volatile. The obtained cycles are a bit too volatile as compared to the NBER.
- One can also use *frequency domain* techniques. The idea is that a (stationary) series can be written as a sum of periodic functions (sine and cosine waves for example):

$$y_t = \sum_i a_i \cos(\omega_i t) + b_i \sin(\omega_i t) \qquad (\star)$$

  The sequence of a_i and b_i gives the loading of the series to cycles of various frequencies. If the a_i and b_i are large for small ω_i, then the series has a lot of low frequencies, meaning that its fluctuations are rather long (periodicity is large). If the a_i and b_i are large for large ω_i, then the series has a lot of high frequencies, meaning that its fluctuations are rather short (periodicity is low). If one is interested in some particular range of periodicity (say 5 to 10 year), the corresponding cycle can be obtained from equation $(\star)$.

PART C : Discussion

BELOW IS an article published in The Economist in December 2021 and entitled "Japan's economy is stronger than many realise". As always in The Economist, the article is not signed. Read carefully this article before answering the questions below. For each question, you should propose a structured answer.

C.1 Write a 10-line summary.

- In the 1990s, Japan was facing slow growth, deflation and a large public debt (230% of GDP) that was considered as unsustainable.
- This was a situation coined as "secular stagnation", i.e. low inflation, low interest rates, low growth and ageing population.
- The high level of public debt has not been a problem for three rasons: interest rates are low, Japan debt is denominated in Yen and the Bank of Japan holds a large fraction of that debt.
- Japan seems to have adapted to the situation. It is true that inflation is still below target, and that monetary policy have been inefficient tu pull the economy out of its mild deflation. But overall growth per capita is comparable to other G7 countries.

- Structural policies that boost innovation and reform the labour market are needed, but somewhat hindered by the Japanese society.

C.2 Explain why the first quote of ADAM SMITH illustrates the functioning of an economy in the Malthusian regime (roughly 500 words).

- In this quote, ADAM SMITH states that what the mark of a country prosperity is gauged nut its population size. This is quite different from the current perspective, according to which a country prosperity is gauged by its level and growth of *income per capita*.
- ADAM SMITH reasoning is very much indicative of an economy in what we call a "Malthusian regime".
- In a Malthusian regime, income per capita is roughly constant, close to subsistence level. Changes in the availability of resources translate into population changes.
- Average growth rate of income per capita has been negligible for a long time : 0.05% a year between 1000 and 1820 (MADDISON). Increases in resources and technology translated into increases in population. World population was 231 millions in 1 AD and 268 millions in 1000 AD, then 438 millions in 1500 AD. Resource expansions after 1500 implied an increase in population growth (1041 millions people in 1820)
- Let us consider technology as a measure of a country prosperity. In the Malthusian regime, a technological improvement will increase population in the long run. But it will not lead to a permanent increase in income per capita.
- One can formally illustrate this mechanism with the CLARK's model we have seen in class. The model has three equations:

  Demography (1): the birth rate b is high when income per capita y is high.

  $$b(t) = b\underbrace{(y(t))}_{+}$$

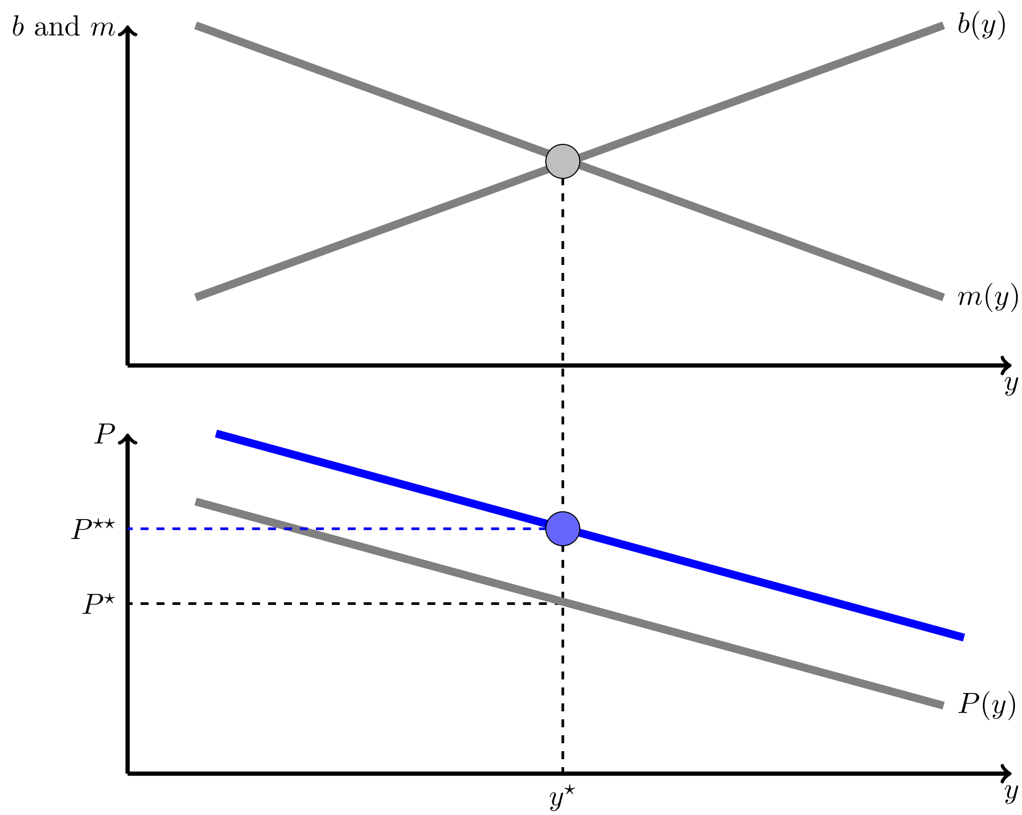
  Demography (2): the mortality rate m is low when income per capita y is high.

  $$m(t) = m\underbrace{(y(t))}_{-}$$

  Technology: Population P is low when income per capita y is high.

  $$P(t) = P\underbrace{(y(t))}_{-}$$

- Graphically, technical progress will shift the production function upwards (the new blue line), which will increase population at the new steady state $P^{\star\star}$, as shown on the figure, but will not change income per capita $y^{\star}$ in the long run.



C.3 Show where the text refers to what we have called in class a "liquidity trap" (to be defined) (roughly 500 words).

- A liquidity trap is a situation in which the nominal interest is at it's lower bound (typically zero) or very close to it. There is slow growth and little inflation or deflation in such a situation.
- Expansionary monetary policy cannot reduce any further the nominal interest rate, and therefore cannot favour consumption and/or investment in the short run.
- Why? Because of the portfolio decision made by economic agents.
- But when the nominal interest rate is at its lower bound, the opportunity cost of money is low (or zero), so that increases in money supply do not increase the supply of loanable funds, and hence does the reduce the nominal interest rate.
- This was shown in class in two models: IS-LM and the micro-founded model of Krugman. (Here it is possible to quickly describe either IS-LM in the liquidity trap or the Krugman model.

– There are good reasons to believe that Japan is in a situation akin to a liquidity trap. As seen on Figure C.1, the nominal interest rate is at zero. (not that the figure shows the 10-year government-bond yield, whereas we generally think of the interbank market rate as the monetary policy instrument)

– The article refers to the inefficiency of monetary policy: "*In 2013 the BOJ's governor, Kuroda Haruhiko, embarked on dramatic monetary easing [...] nor did inflation come near the 2% target.* " and "*Getting inflation up has not proved easy. Under Mr Kuroda, the BOJ expanded quantitative easing, adopted inflation targeting, and purchased a wider variety of assets. That helped pull the country out of its mild deflation, but only barely.*"

– The article also refers to the role of inflation expectations in a liquidity trap: "*Once inflation expectations are anchored around zero, raising them is hard.*". This is something we have studied in the KRUGMAN model. When the economy is in the liquidity trap, monetary policy can recover some effectiveness if it can manage expectations (something we call *forward guidance*). The idea is to convince the economic agents that monetary policy will be lax for a sufficiently long enough period, so that inflation expectations raise, which should increase the nominal interest rate. This is what KRUGMAN refers to as "the central bank committing to be irresponsible". Because the BOJ has established a strong reputation of maintaining price stability in the past, expectations are anchored around zero, and shifting them up is difficult.

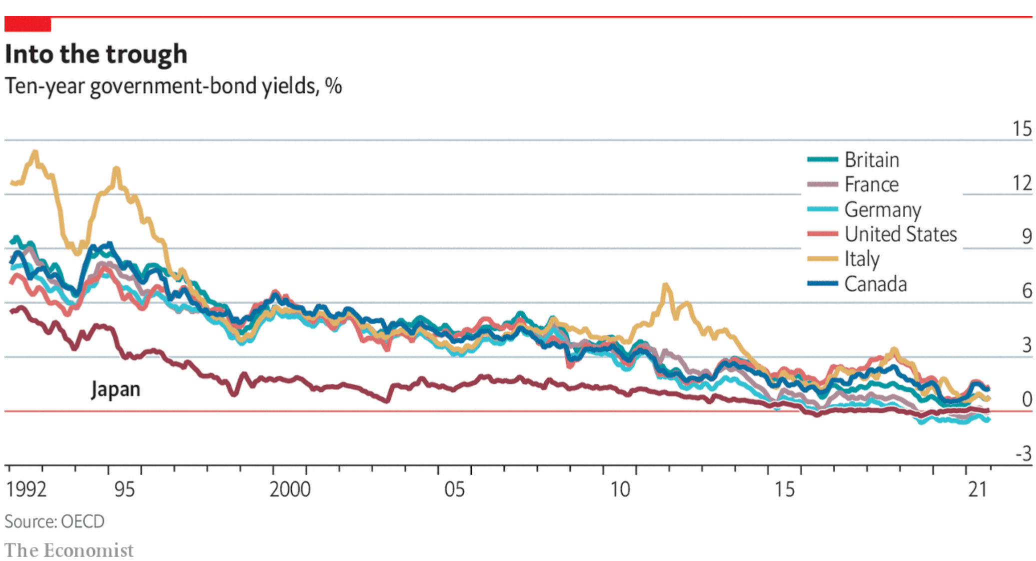
"Japan's economy is stronger than many realise.
Not bad, but could be better "
The Economist, Dec 7th 2021

"The most decisive mark of the prosperity of any country is the increase of the number of its inhabitants," wrote Adam Smith in "The Wealth of Nations" in 1776. Later David Ricardo and Thomas Malthus traded barbs over whether the food supply would keep up. By 1937 John Maynard Keynes was warning of future population decline, with deleterious economic effects.

Japan is the canary in this coal-mine. In the 1980s its booming economy struck fear in the world. After the bubble burst in the 1990s, public debt ballooned and deflation set in. Many in the West said Japan's debt was unsustainable and the Bank of Japan (BOJ) should do more to boost inflation. In 2013 the BOJ's governor, Kuroda Haruhiko, embarked on dramatic monetary easing. The debt hovered around 230% of GDP. A strange thing ensued: no fiscal crisis struck, nor did inflation come near the 2% target. "The standard textbook on macroeconomics needs an additional few chapters–it doesn't capture the problems Japan faced," says Shirakawa Masaaki, Mr Kuroda's predecessor.

Many rich countries now face similar "secular stagnation": low inflation, low interest rates and low growth. Although higher inflation has emerged recently, financial markets suggest secular stagnation will return soon. Demography is a big factor; Japan simply started ageing and shrinking earlier. As Japan has adapted and others have become more like it, some economists are seeing its economy in a new light.

Figure C.1



Debt has not turned out to be such a problem. "What we thought used to be fiscal limits are no longer fiscal limits," argues Adam Posen, of the Peterson Institute for International Economics (PIIE) think-tank. "[Japan] has forced people to confront reality: the interest rate can stay below the growth rate for very long periods." The public debt has been above 100% of GDP for almost 25 years without causing a crisis.

It helps that the country borrows in its own currency, the government has big financial assets and the BOJ holds a big share of the debt. But as David Weinstein of Columbia University argues, Japan has also managed to contain spending. Since 2000, he writes in a paper with Mark Greenan, spending per head on the elderly has actually fallen. "There's a quiet functionality that people miss," says Mr Weinstein. "Markets are sanguine because Japan has a rare ability to adjust." With low marginal tax rates, there is also room to raise revenues.

Some still fear what would happen if interest rates were to go up. The argument that Japan need not worry about its debt because it can respond by levying taxes is "too academic", says Yoshikawa Hiroshi, president of Rissho University. Raising consumption taxes is a political loser. Policymakers are also haunted by the spectre of external shocks that create new fiscal needs. Recently Yano Koji, a vice-minister of finance, caused a stir with a column comparing the country's fiscal situation to the Titanic.

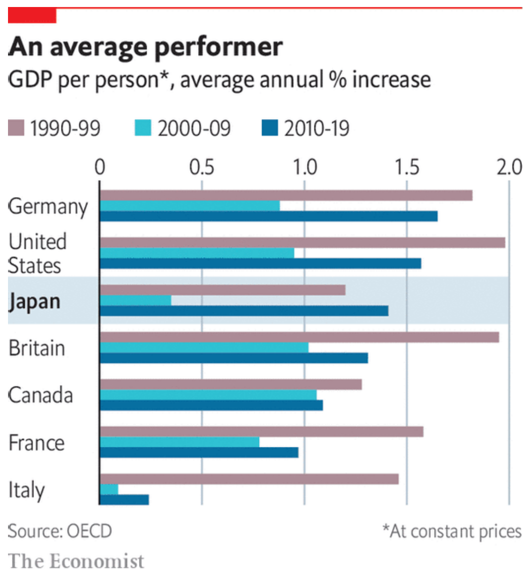
Seeking fiscal space

Yet some reckon fiscal policy ought to be used more forcefully. Many now regret two ill-timed consumption-tax increases in 2014 and 2019. Mr Abe's preferred candidate in the recent LDP leadership race called for postponing the government's primary-balance target until the BOJ hits its inflation target. In late November Mr Kishida's government announced a huge fiscal stimulus worth ¥55.7trn ($483bn).

Getting inflation up has not proved easy. Under Mr Kuroda, the BOJ expanded quantitative easing, adopted inflation targeting, and purchased a wider variety of assets. That helped pull the country out of its mild deflation, but only barely. "We misunderstood the inflation issue," says Mr Posen. "It turns out that secular stagnation is far more real and persistent than we thought."

Once inflation expectations are anchored around zero, raising them is hard. Moreover wages have not risen much, despite a tight labour market. Unions prefer stability of employment to wage rises, reckons Nakaso Hiroshi, a former deputy governor of the BOJ. Companies have gradually taken on more "non-regular" workers on part-time contracts; they account for 40% of the labour force, twice as much as in 1990. Perverse incentives may help depress their wages: many women limit their hours or incomes to secure a tax deduction for married couples earning below a threshold.

Figure C.2



Ageing, shrinking populations may also be weighing on demand and thus inflation. For Mr Shirakawa, that is a vindication of sorts. He argued at the BOJ that deflation was more a symptom of factors causing low growth, not the cause. In a new book, "Tumultuous Times", Mr Shirakawa says the impact of demographic change on growth "is still under-appreciated".

Overall growth has remained sluggish, but growth per head has recently been comparable with others in the G7. Unemployment has been minimal, longevity has increased and inequality has stayed relatively low. "Maybe Western economists who were so critical of Japan circa 2000, myself included, should go to Tokyo and apologise to the emperor," Paul Krugman, an economist, tweeted in 2020. "Not that they did great; but we did much worse."

Yet Japan could still do better. Public spending should be aimed more at improving long-term growth. Economists in Tokyo fret that the government has wasted its pandemic stimulus on handouts: a study by Hoshi Takeo of the University of Tokyo finds that fiscal support in 2020 was more likely to go to companies that were already struggling before covid-19.

Boosting productivity could help to offset the impact of the shrinking population. Mr Yoshikawa reckons that innovation is key to growth, and that ageing creates new problems that entrepreneurs can solve. Generational shifts may help. While many still prefer stable sarariman (salaryman) jobs in big firms, some of today's brightest graduates go into startups. "There is a burgeoning wave–they are a different species," says Mr Niinami. But structural reforms are necessary too, particularly to the inflexible labour market. It should be easier for workers to move between firms and industries, and harder for firms to exploit non-regular workers.

Yet pressure for reforms is lacking, partly due to resistance from vested interests, but also because life remains comfortable enough. “It’s not an acute disease, it’s chronic,” says Mr Nakaso. “You don’t really feel the pain, but it impacts your health in the long-term.” This could be treated. As Mr Posen puts it, “There are 10,000 yen notes lying on the ground waiting to be picked up.” Will Japan’s leaders grab them?